Probability in Statistics

- Introduction (Probability, Sample space & Events)
- Conditional Probability & Baye's Theorem

Probability is the branch of mathematics that deals with the likelihood or chance of different outcomes. Understanding probability helps us make predictions about uncertain events.

- Experiment: An activity or process that leads to one of several possible outcomes.
- Sample Space (S): The set of all possible outcomes of an experiment.
- Event (E): A subset of the sample space. An event occurs if the outcome of the experiment is a member of the event.

Experiment → tossing a coin

Sample space → {H, T}

Event → Probability of getting a head

$$P(E) = \frac{\text{Num of favorable outcomes to } E}{\text{Total num of outcomes}}$$

$$\boxed{P(E) = \frac{1}{2}}$$

Types of Events

① Independent Event

Consider the experiment of rolling a die and flipping a coin. The outcome of the die roll does not affect the outcome of the coin flip.

getting 3

$P(A) = \frac{1}{6}$

getting head

$P(B) = \frac{1}{2}$

Probability of both events occuring

$$P(A \cap B) = P(A) * P(B)$$

$$= \frac{1}{6} * \frac{1}{2}$$

$$= \frac{1}{12}$$

② Dependent Event

Consider the experiment of drawing two cards from a deck without replacement. The outcome of the first draw affects the outcome of the second draw.

ACE

King

Probability of both events occuring

$P(A) = \frac{4}{52}$

$$P(A \cap B) = P(A) * P(B|A)$$

$P(B) = \frac{4}{51}$

King given an ace was drawn first

$$= \frac{4}{52} * \frac{4}{51}$$

$$\approx 0.006$$

(3) Mutually Exclusive Event

↳ two event can't happen simultaneously

Consider the experiment of rolling a die. The events "rolling a 3" and "rolling a 5" are mutually exclusive.

$$P(A) = 1/6$$

$$P(B) = 1/6$$

Probability of both events occurring

$$P(A \cap B) = 0$$

Probability of either event occurring

$$P(A \cup B) = P(A) + P(B)$$

$$= 1/6 + 1/6$$

$$= 2/6 = 1/3$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred. It is a fundamental concept in probability theory and is used to update the probability of an event based on new information.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \; ; \; P(B) \neq 0$$

Probability of A given B event already have taken place

Consider drawing two cards from a standard deck of 52 cards without replacement. What is the probability that the second card drawn is a King, given that the first card drawn was an Ace?

$$P(A) = \frac{4}{52}$$

$$\boxed{P(B|A) = \frac{4}{51}}$$

$$\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}}$$

$$P(A \cap B) = P(B|A) * P(A)$$

$$= \frac{4}{51} * \frac{4}{52}$$

$$\approx 0.006$$

Baye's Theorem

Bayes' Theorem is a fundamental theorem in probability theory that allows us to update the probability of an event based on new evidence. It is named after the Reverend Thomas Bayes and provides a way to revise existing predictions or theories (posterior probabilities) given new or additional evidence.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\boxed{P(A \cap B) = P(A|B) * P(B)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) * P(A)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B) * P(B)}{P(A)}$$

$$P(B) = P(B|A) * P(A) + P(B|\bar{A}) * P(\bar{A})$$

the law of total probability

$P(B_1) = 1/3$ —— $P(R|B_1) = 2/3$

$P(B_2) = 1/3$ —— $P(R|B_2) = 1/3$

$P(B_3) = 1/3$ —— $P(R|B_3) = 3/4$

Imagine you have three boxes:

- Box 1 contains 2 red balls and 1 blue ball.
- Box 2 contains 1 red ball and 2 blue balls.
- Box 3 contains 3 red balls and 1 blue ball.

If you randomly choose a box and then randomly draw a ball from the chosen box, the probability of drawing a red ball can be calculated using the law of total probability.

$$P(R) = P(B_1) * P(R|B_1) + P(B_2) * P(R|B_2) + P(B_3) * P(R|B_3)$$

$$= 1/3 * 2/3 + 1/3 * 1/3 + 1/3 * 3/4$$

$$= 7/12$$

The law of total probability provides a way to break down complex probability calculations into simpler parts.

Let's consider an example of a medical test for a disease.

- Let A be the event that a person has the disease.
- Let B be the event that the person tests positive for the disease.

$$\begin{cases} P(A) = 0.01 \\ P(\bar{A}) = 0.99 \\ P(B|A) = 0.99 \\ P(B|\bar{A}) = 0.05 \end{cases}$$

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$$= \frac{P(B|A) * P(A)}{P(B|A) * P(A) + P(B|\bar{A}) * P(\bar{A})}$$

$$= \frac{0.99 * 0.01}{0.99 * 0.01 + 0.05 * 0.99}$$

$$\approx 0.167$$

Union of Two Events (General case):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Union of Two Events (Mutually Exclusive Events):

$$P(A \cup B) = P(A) + P(B)$$

Difference of Two Events:

$$P(A - B) = P(A) - P(A \cap B)$$

Intersection of Two Events (General case):

$$P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Intersection of Two Events (Independent Events):

$$P(A \cap B) = P(A) \times P(B)$$

Complement of Two Events:

$$P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B)$$